Design of Ai powered Issue Reporting & Resolution system for large campus.

1.Devesh Meshram, 2.Shubham kokate, 3.Tejas Patil, 4.Manali Ghule, 5.Dr. Neerajkumar Sathawane

Department of Computer Science & Business Systems,
JSPM's Rajarshi Shahu College of Engineering,
Pune-411033, India

Abstract

Large corporate and institutional campuses continue to struggle with maintenance management, where complaints logged via email or phone lead to delayed responses, poor tracking, and lack of transparency. To address this, the proposed platform is a web-based AI-powered issue reporting and resolution system built around two portals — an administration dashboard and a campus user portal. The first core feature automates maintenance verification by analyzing camera images of key locations at scheduled intervals, confirming whether tasks have been completed and reflecting results on the admin dashboard in real time. The second feature allows users to report problems by submitting photos, which an ML model automatically categorizes into classes such as cleaning, electrical, plumbing, or infrastructure damage — enabling faster and more organized responses. By replacing manual reporting with intelligent automation, the system improves accountability, reduces response times, and lays the foundation for smart campus management. Future enhancements could include automated task assignment, predictive maintenance scheduling, and IoT integration for real-time facility monitoring.

Keywords: AI-Powered Issue Reporting, Computer Vision, Image Classification, Smart Campus, Deep Learning, Automated Maintenance Verification

1. INTRODUCTION

Large campuses such as corporate parks, universities, hospitals, and industrial facilities face daily operational maintenance challenges requiring continuous monitoring of routine activities including cleaning, waste management, and infrastructure upkeep. Beyond routine tasks, unforeseen issues like electrical faults, plumbing failures, and equipment breakdowns demand timely reporting and resolution.

Most existing systems rely on manual processes — phone calls, emails, or written complaints — resulting in delayed responses, poor tracking, and excessive follow-ups with maintenance teams. The lack of a structured system makes it difficult for administrators to manage and prioritize issues effectively.

Advancements in Artificial Intelligence, Machine Learning, and Computer Vision now offer the potential to significantly enhance facility management. These technologies can automate maintenance tracking, enable image-based problem identification, and streamline communication between campus users and administrators, minimizing manual intervention and ensuring greater transparency.

Computer vision models have proven effective in waste detection, cleanliness assessment, and infrastructure damage identification, while ML-based reporting systems can automatically classify submitted complaints by issue type. This survey consolidates existing research across AI-based issue reporting systems, smart campus management platforms, and automated maintenance monitoring technologies — identifying current solutions, their limitations, and research gaps — to lay the foundation for

building more efficient, AI-driven campus maintenance systems.

2. AIM & OBJECTIVES

Aim of the Project

To design and develop an AI-powered web-based issue reporting and resolution system for large campuses that integrates automated camera-based daily maintenance verification with machine learning driven issue classification, enabling efficient, transparent, and data-driven facility management through a dual-portal platform accessible to both campus individuals and administrators.

Objectives of the Project

1. To implement an automated camera-based daily maintenance verification system that captures and compares site images at scheduled intervals using computer vision techniques such as image similarity analysis and scene change detection, in order to determine whether daily maintenance tasks have been completed and reflect verified results on the administrative dashboard without manual intervention.

2. To develop a machine learning model for intelligent issue classification and image analysis that processes user-submitted photographs and automatically categorizes reported problems into predefined classes such as electrical faults, plumbing failures, structural damage, and cleanliness violations, thereby minimizing manual triaging effort and improving detection accuracy.

3. To design and develop a secure dual-portal web platform with role-based access control, comprising a campus individual portal for issue submission and real-time status tracking, and an administrator portal for maintenance oversight, AI-generated report viewing, and team management, ensuring a structured and accountable workflow across all stakeholders.

4. To implement an intelligent task assignment and priority management mechanism that automatically routes reported issues to the appropriate maintenance team based on issue category, urgency level, and location, while maintaining a clearly defined resolution lifecycle from submission through to closure.

5. To develop a real-time analytics and performance monitoring dashboard that provides administrators with actionable insights including issue frequency trends, category-wise distribution, maintenance compliance rates, and average resolution times, supporting informed and data-driven facility management decisions.

6. To evaluate the scalability and adaptability of the proposed system architecture across diverse large-scale environments including corporate campuses, educational institutions, and hospitals, demonstrating its potential as a replicable and extensible solution for smart infrastructure management.

3. LITERATURE SURVEY

Research from Indian institutions has made significant contributions to AI-powered campus platforms directly relevant to the proposed system. Sameera et al. [1] proposed the Smart Campus Grievance Management System using the MERN stack with NLP-based priority detection, achieving 96.2% routing accuracy and a System Usability Score of 91/100, proving that role-based dashboards with automated escalation significantly outperform manual grievance handling. Chaudhari et al. [2] developed an AI-powered Campus Assistant at Vishwakarma University, Pune, using LangChain and Retrieval-Augmented Generation, reducing query resolution time from 24–48 hours to 3.24 seconds with 94.3% chatbot accuracy, establishing a strong precedent for unified AI-driven campus platforms in Indian higher education. Sowmya et al. [3] validated camera-based campus monitoring feasibility through a Raspberry Pi-based Smart Campus Bot integrating a Pi Camera, RFID verification, and IoT sensors, achieving 85% facial detection accuracy, directly informing the camera integration approach of the proposed system. On the international front, Pongpaichet et al. [4] developed the Smart Complaint Management System using Sequential Minimal Optimization with TF-IDF

Sr. No.	Ref.	Author(s) & Year	Institution / Country	Key Technology	Core Feature	Limitation
1	[1]	Sameera et al. (2025)	GMR Institute of Technology, India	MERN Stack, NLP, MongoDB	Automated complaint routing, priority detection, escalation	No image detection, no camera verification
2	[2]	Chaudhari et al. (2025)	Vishwakarma University, Pune, India	LangChain, RAG, ChatGroq, FAISS, React	AI chatbot, admin panel, unified campus platform	No facility issue reporting or maintenance verification
3	[3]	Sowmya et al. (2025)	VVIET, Mysore, India	Raspberry Pi, Pi Camera, RFID, OpenCV	Camera monitoring, facial recognition, IoT integration	Hardware-dependent, no web portal or complaint management
4	[4]	Pongpaichet et al. (2019)	Mahidol University, Thailand	SMO, TF-IDF, Mobile App, Chatbot	ML complaint classification, duplicate detection, status tracking	Text-only input, no image or camera-based detection
5	[5]	Mohamed et al. (2025)	Jamhuriya University, Somalia	Multinomial Naive Bayes, TF-IDF, NLP	94% classification accuracy, admin analytics dashboard	No visual input, text-only classification
6	[6]	Zhou et al. (2025)	Chongqing College, China	TensorFlow, Facial Recognition, Spring Boot, Vue.js	AI campus management, access control, occupancy monitoring	No issue reporting portal, no maintenance verification

feature extraction achieving an F-Measure of 0.796 across complaint categories, while also introducing duplicate complaint detection through keyword similarity matching.

[5] further achieved 94% classification accuracy using Multinomial Naive Bayes combined with TF-IDF across Academic, Finance, and Equipment categories with a real-time administrative analytics dashboard. Zhou et al. [6] presented a comprehensive AI campus management system with facial recognition achieving 99.12% accuracy and 99.97% system availability under stress testing, establishing key performance benchmarks for large-scale deployments. Despite these advances, a critical research gap persists across all reviewed works — no existing system combines automated camera-based daily maintenance verification with ML-powered image classification of user-reported infrastructure issues within a unified dual-portal web platform, which forms the core novelty and motivation of the proposed system.

4. EXISTING SYSTEMS

The management of campus infrastructure and complaint resolution has traditionally relied on manual, paper-based processes involving physical complaint forms, verbal communication, and in-person reporting to administrative offices. These conventional methods suffer from significant drawbacks including delayed response times, lack of transparency, loss of complaint records, absence of real-time tracking, and heavy dependence on human intervention at every stage of the resolution process. As digital transformation accelerated across educational and corporate institutions, several web-based and AI-assisted systems emerged to address these inefficiencies, each contributing meaningful advancements while simultaneously revealing new limitations.

Early digital complaint systems replaced paper forms with basic online portals that allowed users to submit grievances electronically. While these platforms improved accessibility and reduced paperwork, they continued to rely on manual complaint routing, static predefined categories, and human administrators for

classification and assignment. These systems lacked any form of automation, intelligent prioritization, or real-time notification, making them only marginally better than their physical counterparts in terms of actual resolution efficiency. The absence of duplicate complaint detection further resulted in redundant submissions that increased administrative workload rather than reducing it.

The introduction of machine learning and natural language processing into complaint management marked a significant advancement. Systems began automatically classifying complaints into relevant categories and routing them to appropriate departments without manual intervention. Text-based classification models using algorithms such as Support Vector Machines, Naive Bayes, and Decision Trees demonstrated promising accuracy in categorizing complaints. However, these systems remained entirely dependent on textual input and lacked the capability to process or analyze images submitted alongside complaints. In real-world infrastructure problems such as water leakage, structural cracks, electrical faults, or cleanliness violations, visual evidence is often the most critical form of information, and its absence severely limited the diagnostic capability of these platforms.

On the monitoring side, existing campus management systems have explored camera-based surveillance and IoT sensor integration for security and access control purposes. Facial recognition systems, infrared occupancy detectors, and CCTV networks have been deployed in various institutions to automate entry management and monitor footfall. However, these camera deployments serve exclusively reactive security functions and are not configured to assess the completion status of maintenance tasks or verify whether scheduled cleaning, repair, or upkeep activities have been carried out. The concept of using scheduled image capture and computer vision-based comparison to proactively determine maintenance compliance remains entirely absent from existing deployments.

Furthermore, current systems operate in isolation with no integration between user-reported complaint management and automated visual monitoring. Complaint portals and surveillance systems exist as completely separate platforms with different administrative interfaces, data pipelines, and resolution workflows. This fragmentation creates operational blind spots where routine maintenance failures go undetected until physically reported, and where reported issues lack visual validation to assist in accurate classification and priority assignment. The absence of a unified platform that consolidates both proactive automated monitoring and reactive user-driven reporting represents the most significant gap in the current landscape of campus facility management systems.

Critical Gap: Although text-based complaint classification and camera-based campus monitoring have advanced independently, no existing system integrates both within a single unified platform. Current systems are entirely reactive, relying solely on text descriptions with no ability to process visual evidence or proactively verify daily maintenance compliance through image capture and computer vision. The absence of a cohesive platform combining automated camera-based maintenance verification, image-based ML classification, and dual-portal complaint management represents the most critical gap in existing literature.

4. TECHNOLOGY STACK

Component	**Technology Used**
Frontend	React.js
Backend	Node.js, Express.js
Database	MongoDB
ML Model	Python, TensorFlow / Keras, CNN (ResNet50 / MobileNetV2)
Image Comparison	OpenCV, SSIM (Structural Similarity Index Measure)
Authentication	JSON Web Tokens (JWT)
API Integration	Flask (Python–Node.js bridge)
Notifications	Email / In-App Alerts
Deployment	Cloud-Based (AWS / Heroku)

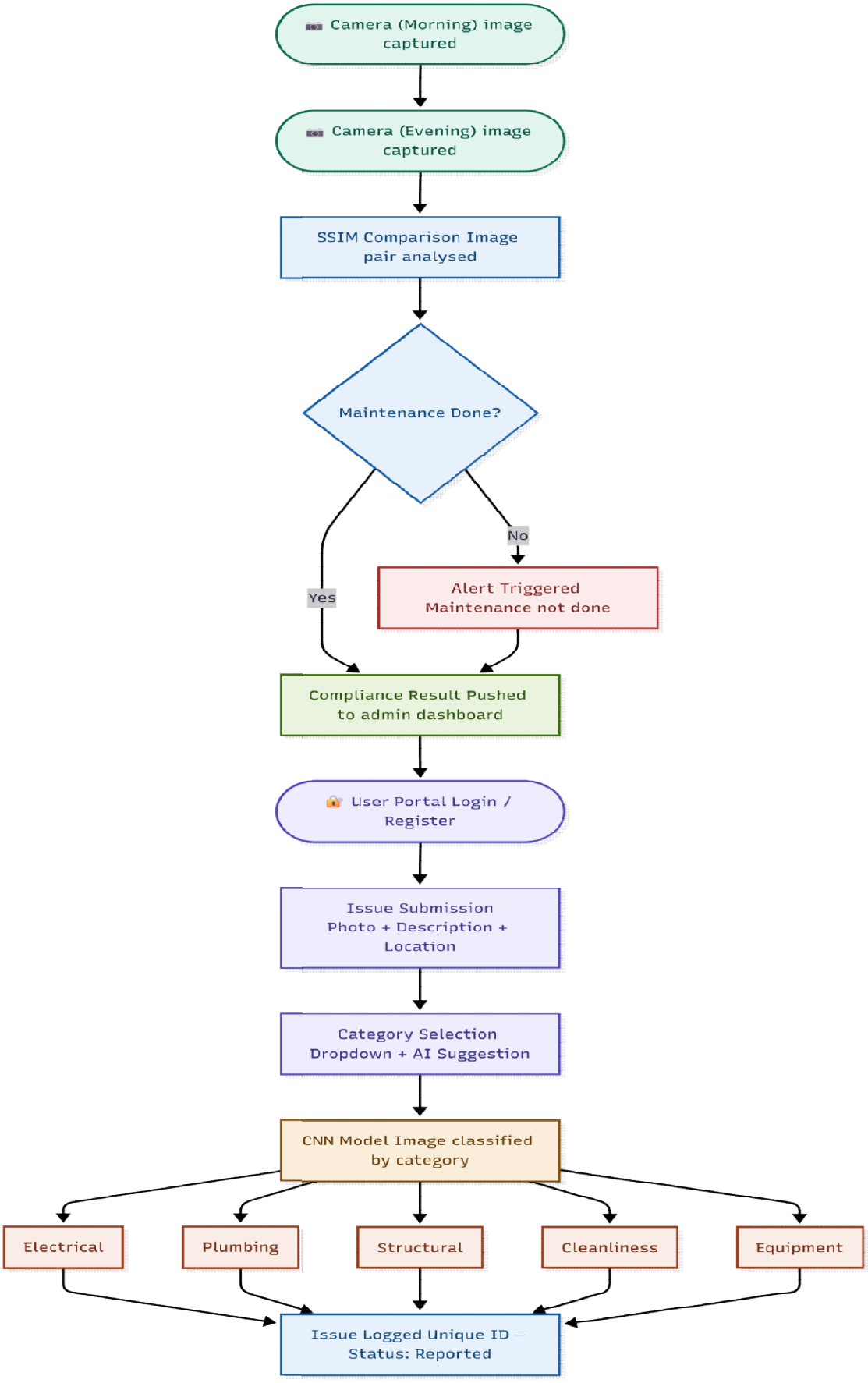


Fig. System Workflow

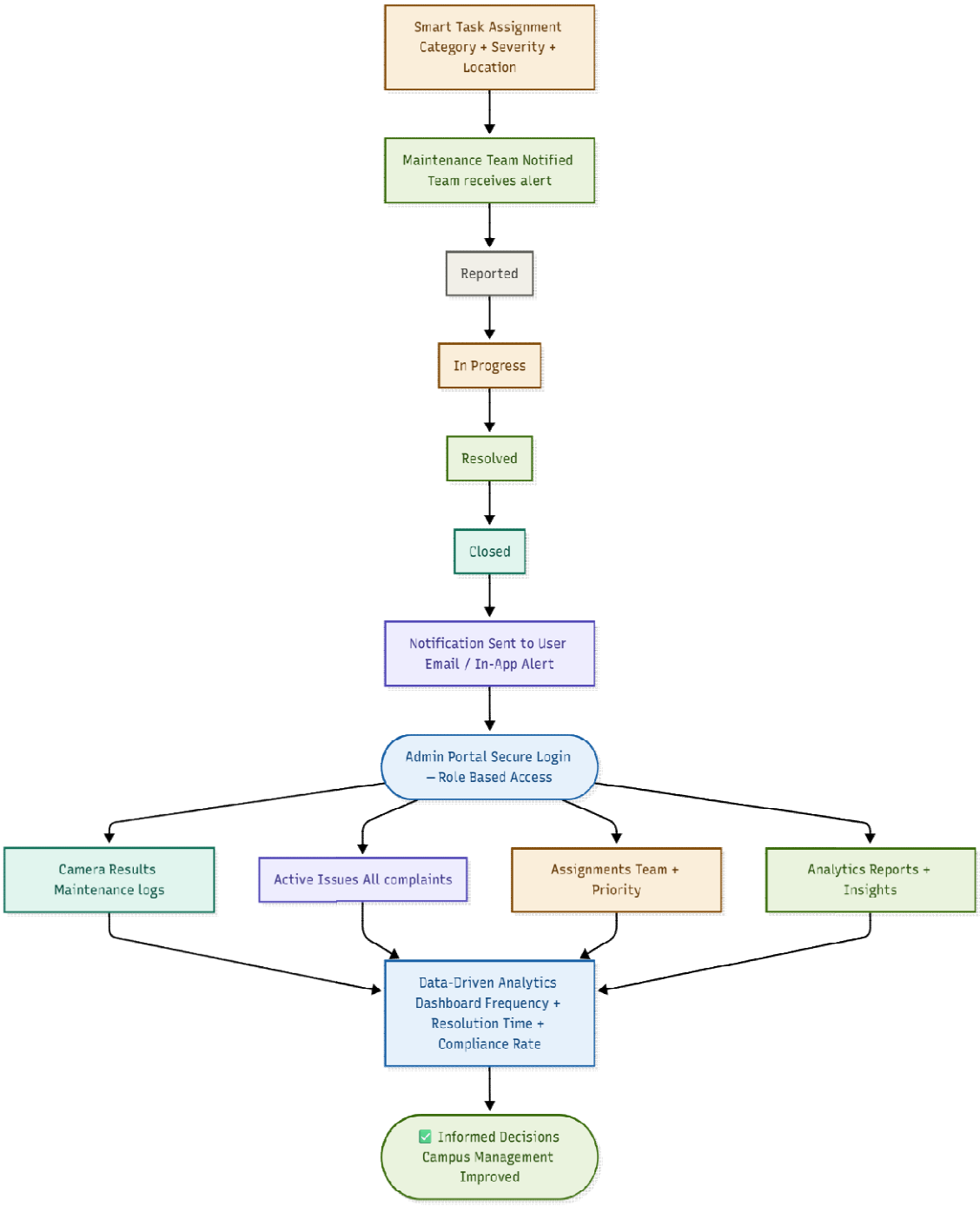


Fig. System Architecture

5. PROPOSED SYSTEM

The proposed AI-Powered Issue Reporting and Resolution System is a unified web-based platform designed to address the critical limitations of existing campus facility management systems by integrating two parallel functional pipelines within a single cohesive architecture. The first pipeline addresses proactive maintenance monitoring through an automated camera module that captures morning and evening photographs at designated campus zones and processes them through a computer vision pipeline using the Structural Similarity Index Measure to automatically determine whether scheduled daily maintenance tasks have been completed. The result is pushed directly to the administrator dashboard as a compliance confirmation or non-compliance alert, eliminating dependency on manual inspection entirely. The second pipeline addresses reactive issue reporting through a user-facing complaint portal where campus individuals submit infrastructure problems with photographic evidence, which is simultaneously processed by a Convolutional Neural Network based classification model to automatically predict the issue category from five predefined classes — Electrical, Plumbing, Structural Damage, Cleanliness, and Equipment Failure. Both pipelines feed into a unified administrative dashboard providing real-time visibility and data-driven decision-making capabilities to campus management.

The system provides two distinct role-based portals secured through JSON Web Token authentication. The Campus Individual Portal enables registered users to submit issue reports with photographs, location details, and text descriptions, track their complaint status in real time, and receive automated notifications at every stage of the resolution process. The Administrator Portal provides a centralized dashboard consolidating outputs from both the camera module and the complaint management module, allowing administrators to view maintenance compliance results, monitor all active issues, manage team assignments, and access performance analytics from a single unified interface. This dual-portal architecture eliminates the operational fragmentation present in existing systems where complaint portals and monitoring systems operate in complete isolation from one another.

Once an issue is classified and logged, the system automatically assigns it to the appropriate maintenance team based on issue category, campus location, and severity level derived from the visual content of the submitted image. Every reported issue follows a clearly defined four-stage resolution lifecycle — Reported, In Progress, Resolved, and Closed — with each status change automatically triggering a notification to the reporting user, providing complete transparency throughout the resolution process without any manual follow-up communication. The administrator dashboard further incorporates a real-time analytics module providing insights including issue frequency by category, average resolution time, maintenance compliance rate, team performance metrics, and campus hotspot mapping, enabling campus management to transition from reactive problem-solving toward proactive strategic planning and continuous improvement of facility operations.

6. TOOLS & TECHNOLOGIES

The proposed system is developed using a carefully selected set of modern tools and technologies spanning frontend development, backend processing, database management, machine learning, computer vision, and deployment infrastructure, ensuring every component is supported by robust and widely adopted frameworks.

6.1 Frontend Technologies

The frontend is developed using React.js, a component-based JavaScript library that enables the development of two distinct role-based portals within a single modular codebase. The user interface is styled using Tailwind CSS ensuring a clean and responsive design across all device sizes, while Axios handles all HTTP communication between the frontend and the backend server.

6.2 Backend Technologies

The backend is built on Node.js with Express.js serving as the web application framework for managing all REST API endpoints. User authentication and role-based access control are implemented using JSON Web Tokens providing a stateless and secure mechanism for enforcing portal-level access restrictions. Multer is used as middleware for handling

image uploads submitted through the complaint portal.

6.3 Database Technologies

MongoDB serves as the primary database, chosen for its flexible document-oriented schema that accommodates the varied data structures of issue reports, maintenance logs, user profiles, and notification records. Mongoose is used as the Object Data Modeling library on top of MongoDB providing schema validation, data consistency enforcement, and simplified query operations within the backend environment.

6.4 Machine Learning and AI Technologies

The image classification model is developed using Python with TensorFlow and Keras as the core deep learning frameworks. The model employs transfer learning on ResNet50 or MobileNetV2, fine-tuned on a domain-specific dataset of campus infrastructure images. NumPy and Pandas support data processing during training, while Flask exposes the trained model as a REST API endpoint enabling seamless integration between the Python-based AI layer and the Node.js backend.

6.5 Computer Vision Technologies

The maintenance verification pipeline is implemented using OpenCV for image preprocessing including grayscale conversion and Gaussian blur filtering. The Structural Similarity Index Measure is computed using the scikit-image library to quantitatively compare morning and evening image pairs and generate a compliance score. Pillow supports image format handling within the verification pipeline.

6.6 Development and Deployment Tools

Visual Studio Code serves as the primary development environment with Postman used for API testing and validation. Git and GitHub manage version control and collaborative development throughout the project lifecycle. The application is deployed on cloud infrastructure using Heroku or AWS, with Docker optionally used for containerized and portable deployment across environments.

7. CONCLUSION

This paper presented the design of an AI-Powered Issue Reporting and Resolution System for large campuses, integrating automated camera-based daily maintenance verification using the Structural Similarity Index Measure with a CNN-based image classification model for user-reported infrastructure issues, administered through a unified dual-portal web platform. Unlike existing systems that address complaint management and visual monitoring in isolation, the proposed system consolidates both proactive and reactive facility management capabilities into a single cohesive solution supported by intelligent task assignment, real-time status tracking, and a data-driven analytics dashboard.

The proposed system represents a novel contribution to smart campus infrastructure management by being the first to combine visual maintenance verification with image-based issue classification within a unified platform. Its scalable and modular architecture ensures adaptability across corporate campuses, educational institutions, and hospitals. Future work will focus on expanding issue categories, incorporating predictive maintenance using historical data, integrating IoT sensors for enhanced anomaly detection, and developing a dedicated mobile application for improved accessibility.

8. FUTURE SCOPE

Although the proposed system addresses the core requirements of campus facility management, several directions exist for future enhancement. The camera-based verification module can be extended by integrating IoT sensors for continuous real-time anomaly detection beyond scheduled intervals. The CNN classification model can be trained on larger and more diverse datasets to improve accuracy and cover a broader range of issue categories. Predictive maintenance capabilities can be incorporated by analyzing historical issue data to proactively identify campus zones likely to develop infrastructure problems before they are reported. A dedicated mobile application for

Android and iOS can significantly improve accessibility and user adoption, while NLP-based chatbot integration can simplify the issue submission process through conversational reporting. Finally, blockchain-based audit trails and ERP system integration can further strengthen data integrity and operational management across large-scale campus environments.

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